

torch.nn.functional.log_softmax#

[https://pytorch.org/docs/stable/generated/torch.nn.functional.log_softmax.ht](https://pytorch.org/docs/stable/generated/torch.nn.functional.log_softmax.html)

Description:

Apply a softmax followed by a logarithm. While mathematically equivalent to $\log(\text{softmax}(x))$, doing these two operations separately is slower and numerically unstable. This function uses an alternative formulation to compute the output and gradient correctly. See `LogSoftmax` for more details. `input` (Tensor) – input `dim` (int) – A dimension along which `log_softmax` will be computed.

Function Signature

```
torch.nn.functional.log_softmax(input, dim=None,
                                _stacklevel=3, dtype=None)[source]#
```

Parameters

Return type

Returns:

Tensor

Detailed Documentation

Documentation

Apply a softmax followed by a logarithm.

While mathematically equivalent to $\log(\text{softmax}(x))$, doing these two operations separately is slower and numerically unstable. This function uses an alternative formulation to compute the output and gradient correctly.

See `LogSoftmax` for more details.

`input (Tensor)` – input

`dim (int)` – A dimension along which `log_softmax` will be computed.

`dtype (torch.dtype, optional)` – the desired data type of returned tensor. If specified, the input tensor is cast to `dtype` before the operation is performed. This is useful for preventing data type overflows. Default: `None`.